FELIPE SANTOS

Physical AI | Edge AI | AIoT Platforms | Solutions Architect · Tech Lead · Forward Deployed Engineer · Specialist | M.Sc. in AI applied to IoT Cybersecurity

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EXECUTIVE SUMMARY

Senior technology specialist with 8+ years architecting intelligent systems that connect the physical and digital worlds — Physical AI, Edge AI, AIoT Platforms, Computer Vision, Robotics, and Agentic AI Platforms with LLM reasoning cores. Forward Deployed Engineer (FDE) and Practice Lead for Edge AI & Platforms from the Physical AI Studio at Globant, and member of the studio management team. Proven record leading cross-functional teams across AWS, Azure, and GCP, delivering mission-critical platforms for global brands (LATAM Airlines, NBA/LA Clippers, Universal Parks, Disney Parks). Trusted client-facing solutions architect embedding directly with enterprise stakeholders from discovery to production — driving presales, PoCs, MVPs, and multi-project account growth. AI-native engineering practitioner: agentic coding workflows, spec-driven development, context engineering, and AI agent harness design for record-speed delivery. Mentor to multiple engineers, technical hiring gatekeeper, and multilingual communicator (EN/PT/ES).

CORE COMPETENCIES

AI & ML: Physical AI · Edge AI · Agentic AI & Autonomous Systems · LLM Reasoning & Multi-Agent Orchestration · GenAI Platform Architecture · Computer Vision (YOLO, OpenCV, NVIDIA DeepStream, TensorRT, Triton, custom model training) · MLOps (Edge-to-Cloud Pipelines) · PyTorch · TensorFlow · Sensor Fusion · Reinforcement Learning · Real-Time Inference Optimization

IoT & Edge: AIoT Platforms · IoT Device Management at Scale (20,000+ devices) · Azure IoT Hub/Edge · AWS IoT Core · OTA Updates · Zero-Touch Provisioning · MQTT/Mosquitto · BLE Beacons & Gateways · NVIDIA Jetson · Raspberry Pi · Embedded Linux · Robotics · Digital Twins · Industrial Edge Computing · IoT Cybersecurity

Cloud & Platform: Multi-Cloud (AWS, Azure, GCP) · CloudOps · Terraform/IaC · Docker · Kubernetes · CI/CD · Microservices & Event-Driven Architecture · Serverless (Cloud Run, Lambda, ECS, VMs, etc) · High-Throughput Data Ingestion (Pub/Sub, Dataflow, Kinesis, etc) · Go · Python · Bash · Rust · SQL/NoSQL Databases

Leadership & AI-Native Delivery: Forward Deployed Engineering (FDE) · Solutions Architecture · Technical Presales · Technical Roadmaps · Team Leadership · Technical Management · Executive Demos · Clients/Vendors Technical Facing and Focal Point · Mentoring · Technical Hiring · Agentic Coding (Claude Code, Codex, OpenCode, etc) · Spec-Driven Development · Context & Token Engineering · Agent Harness Design

PROFESSIONAL EXPERIENCE

GLOBANT — Tech Lead · Solutions Architect · Physical AI FDE · Technology Specialist

May 2022 – Present

Physical AI Studio — Edge AI, AIoT, Computer Vision, and Robotics solutions for aviation, sports & entertainment, smart venues, utilities, and manufacturing.

Studio & Technical Leadership

- Forward Deployed Engineer and lead for Edge AI and Platforms verticals within a 50+ person Physical AI Studio; also contributor to Robotics and Embedded practices.
- Member of the studio management team: drive hackathons, tech sessions, internal R&D prototypes, and staffing strategy for studio projects.
- Tech Lead of internal product initiatives: RobOps platform and intelligent edge devices for human-robot interaction (Edge AI + robotics prototypes).
- Solutions Architect in presales: technical pitches, solution design, PoCs, MVPs, and live demos contributing to multiple six-figure USD project wins, plus account extensions and upsells.
- Mentor to multiple engineers and studio hiring gatekeeper: 10+ technical interviews for staffing candidates.

LATAM Airlines — Physical AI FDE & Lead Architect, Agentic AI Operations Platform

2026 – Present

- Embedded directly with airline operations and technical stakeholders as Forward Deployed Engineer (FDE) and Principal Architect of an AI platform with an LLM reasoning core for real-time monitoring, management, and control of airport turnaround operations — anomaly detection, operational risk identification, and autonomous decision-making.

- Designed and delivered event-driven GCP architecture (Compute Engine, Pub/Sub, Cloud Run APIs, Dataflow, Firestore) in Go and Python, leading end-to-end integration and multi-source sensor fusion across computer vision vendors, airline subsystems, and a proprietary BLE beacon/gateway solution for people and asset tracking.

Universal Parks — Jurassic Park Ride | Tech Lead, Architect & Principal Engineer 2025 – 2026

- Architected a real-time computer vision safety and operational-efficiency platform at the edge: left-object detection, people counting, restricted-zone monitoring, and ride dispatch metrics, integrated with operator alert panels.
- Achieved 0.95+ F1-score with <500 ms end-to-end latency using a custom-trained YOLO model on NVIDIA DeepStream with optimized edge inference (industrial PCs, MQTT/Mosquitto, Go, AWS, Docker, CI/CD); led a team of 5 engineers with on-site demos and client-facing delivery.

BBVA Argentina — Innovation Consultant (Internal Hackathon) 2025

- Advised the team on an eye-tracking applied-intelligence and cybersecurity solution for the bank's mobile application.

Compass (Cosan) — Tech Lead, Smart Metering AIoT Platform 2024 – 2025

- Led a team of 5 building an IoT platform to integrate and remotely manage millions of smart meters: automated device provisioning, device management, security, and OTA updates.
- Engineered high-throughput IoT data ingestion and preprocessing on AWS (IoT Core, Kinesis Firehose, Lambda, ECS, DynamoDB, S3, MQTT) with Terraform IaC in Go/Python; integrated with the data organization for advanced analytics and data intelligence.

LA Clippers (NBA) — Intuit Dome | Senior Engineer promoted to Tech Lead 2023 – 2024

- Built and led delivery of an enterprise device management platform for 20,000+ devices at the Intuit Dome smart venue — remote control, fleet monitoring, OTA updates, zero-touch provisioning, cybersecurity, and automated deployments.
- Led the IoT Edge and IoT Platform engineering teams on the full Azure IoT ecosystem (IoT Hub, IoT Edge) with Terraform, Docker, Python, and CloudOps; promoted from engineer to tech lead during the engagement; on-site delivery and client strategy participation.

HME — Edge-to-Cloud MLOps Platform Engineer 2023

- Developed an end-to-end edge-to-cloud MLOps platform for computer vision models on the NVIDIA stack (DeepStream, YOLO) with Python, Docker, and Bash.

Disney Parks — MLOps Platform Engineer 2022 – 2023

- Built the end-to-end MLOps platform (deployment, monitoring, data consumption, training, and inference) for a computer vision solution improving ride operational efficiency — AWS SageMaker, DynamoDB, Terraform, Python, CI/CD.

BRPHOTONICS — Automation Analyst promoted to Automation & Control Engineer Apr 2020 – Apr 2022

R&D and manufacturing of optoelectronic devices (lasers, amplifiers) for optical communications.

- Designed and fully automated manufacturing, testing, calibration, and quality processes for lasers, amplifiers, and other optoelectronic devices; architected pilot production lines from R&D approval to manufacturing scale.
- Led junior engineers, interns, and technicians across automation, test, and quality engineering; time-series and statistical data analysis for product performance validation.

IDEA! ELECTRONIC SYSTEMS — Product Engineering Intern Jul 2019 – Mar 2020

- Automated calibration, testing, and quality processes for 5G infrastructure optoelectronic devices.

EDUCATION & RESEARCH

M.Sc. in Computer Engineering — UNICAMP (University of Campinas), Brazil 2024 – 2026

- Thesis: Reinforcement Learning applied to cybersecurity in IoT device networks — designed and built an RL agent training framework for autonomous cyber-attack response using the military Kill Chain taxonomy.
- Paper under review at Elsevier *Internet of Things*: view paper (SSRN)
- Collaborated on graduate research projects in digital twins, advanced data analytics, robotics, and AI.

B.Sc. in Electrical Engineering (Control & Automation) — UFPB, Brazil 2014 – 2020

- CNPq/PIBIC research fellow (2016–2019): ultrasonic anemometry for wind velocity measurement; member of the Applied Computational Intelligence group; 4 peer-reviewed publications (IEEE, IOP): IEEE (2018) · IEEE (2019) · IOP J. Phys. (2018)
- Science Without Borders scholar (CAPES): exchange year in Electrical & Computer Engineering at the University of Wisconsin –Milwaukee, USA (2015–2016); Assistant Researcher at Mercer University, USA (2016) — wireless power transfer via resonant inverters.

Languages: Portuguese (native) · English (full professional) · Spanish (full professional)